VF Asia Footwear Laboratory Testing Manual

Test Method for Slide Fasteners to Repeated Opening and Closing	Test method # VF- B26 Page 1 of 7
Date Issued : May,2019 Revision # : 1 Reason for revision : Rectify the unit of measure from kg to kgf for strength of material.	Created by : Nila Babia Erazo

PURPOSE AND SCOPE: To determine the resistance of a slide fastener to repeated opening and closing at specified number of cycles while under tensions.

REFERENCES: SATRA PM50 / BS 3084-2004

EQUIPMENT AND MATERIAL:

1. Test machine in compliance with SATRA PM 50 or BS 3084-2004.
2. Protractor or any device for measuring angles of 30 ± 5° and 60 ± 10°
3. Steel ruler or caliper
4. All type of zippers

SAMPLE PREPARATION:

1. Condition the specimens at 23 ± 2 ℃ Temperature and 55 ± 10% relative humidity for 24 hours prior to test.
2. Specimens shall be taken from each lot randomly.

3. Take a specimen of no less than 200mm in length. If the test fastener is shorter than the minimum distance between the two clamps, then sew strips of either leather or fabric, minimum width of 25 mm, to the closed ends of both stringers. The exact length of the strips will depend on the clamping system of the machine being used.
3. Draw two reference lines 5mm from the ends of the elements on both sides respectively as shown below.

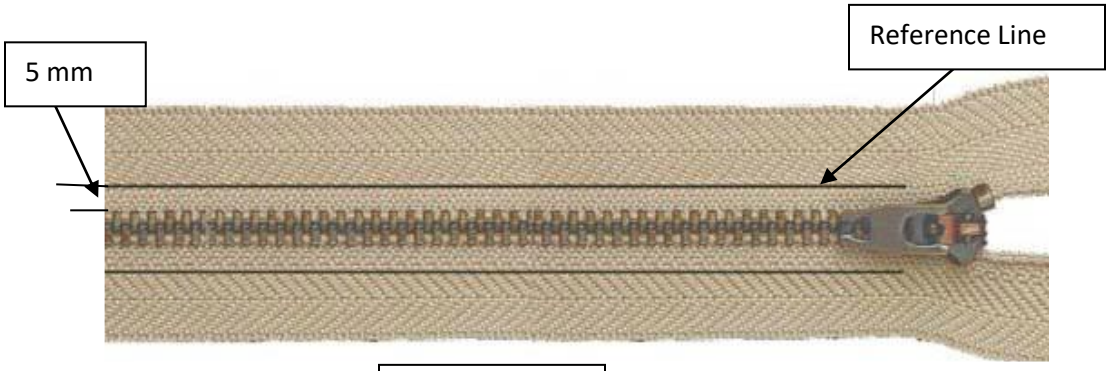


Figure 1

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REPETITONS: 2 specimens per sample

TEST PROCEDURE:

Test should be conducted at standard laboratory conditions @ 23.0±2.0˚C temperature and 55 ±10 % Relative Humidity.

PROCEDURE:

Specimen clamping:

1. Connect a required power supply to the machine and turn on the power switch.
2. Steps on clamping of specimens :

i) Adjust the manual knob until the puller clamping device is in the lowest position and then the indicating lamp will light up.

ii) Attach one or two locating pieces of proper thickness to the puller, and then use a hex key to release the setscrew of the clamping device. Fix the puller under the presser and then lock the setscrew. Refer to the following steps on figures 2-4 .

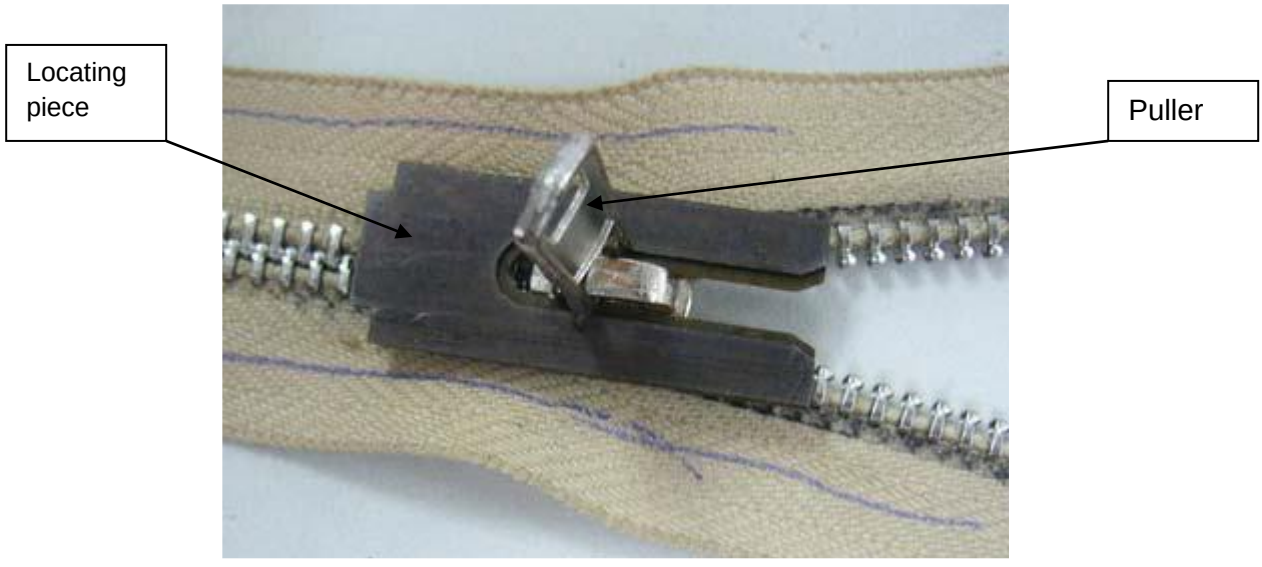


Figure 2

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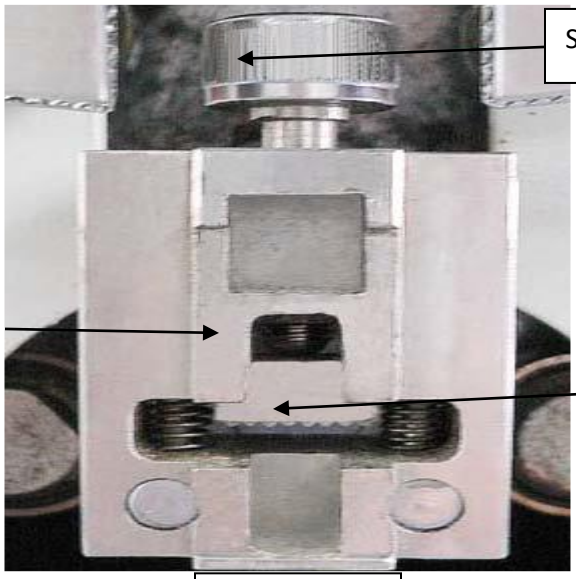


Figure 3

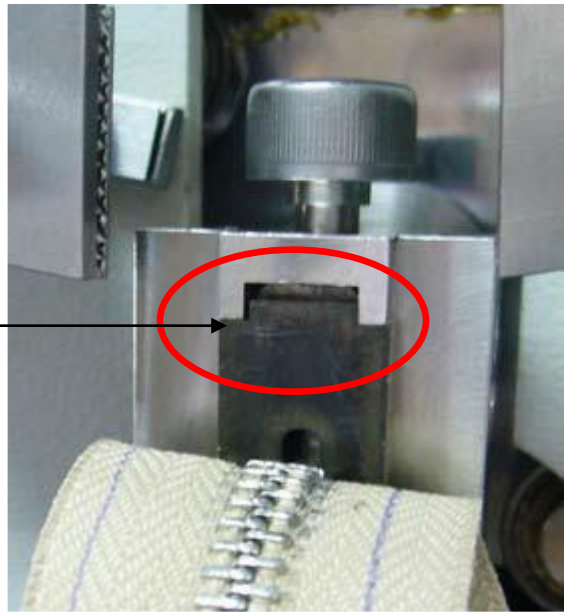
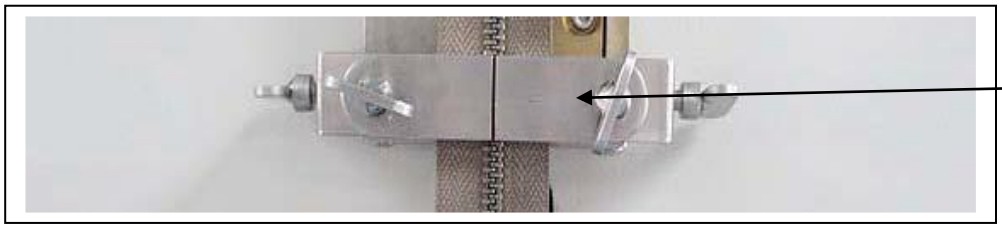


Figure 4

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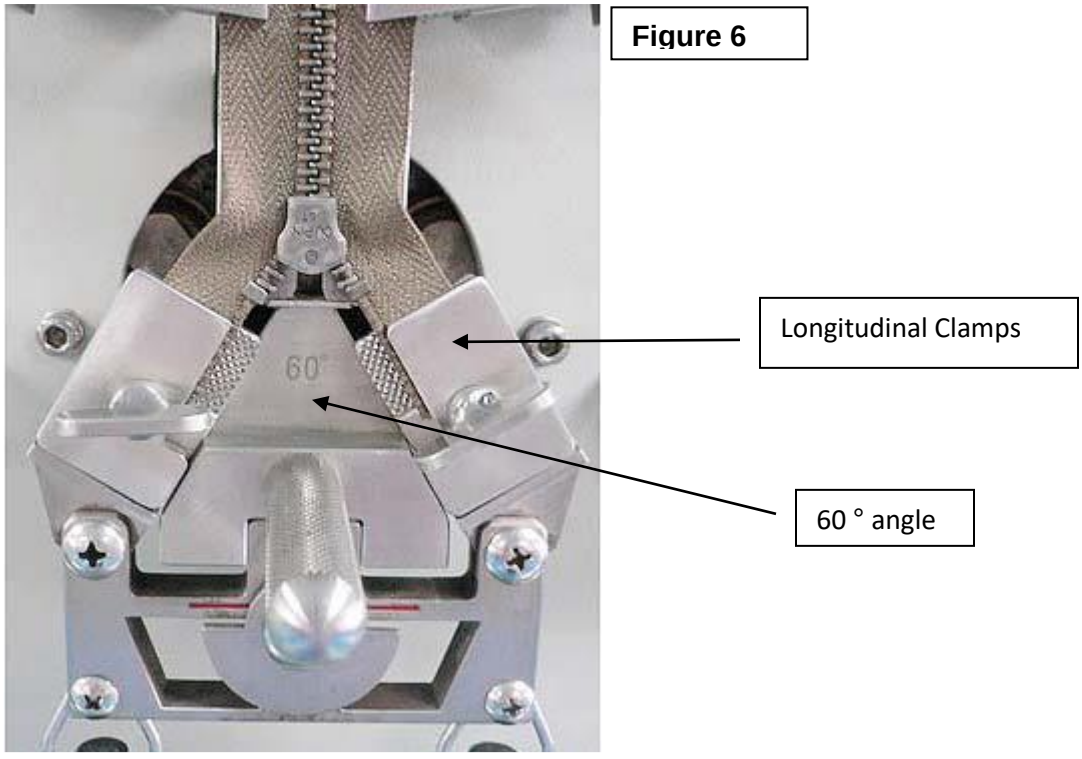
iii) Fix the bottom ends of the zipper (bottom stop) to the upper clamp A in such a way that the central line of the jaws is perpendicular to that of the zipper. Refer to figure 5 below.

Figure 5



iv). Fix the top ends of the zipper (top stop) to the longitudinal clamps, and then use the 60°angle gauge to measure the angle α when the specimen is closed. Refer to figure 6 below.

Figure 6



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v) Remove the 60°angle gauge and grasp the puller to move the zipper upwards until the stop pin was met. Simultaneously check if the red line marked on the stop pin is parallel with the swivel plate. Refer to figure 7 below.

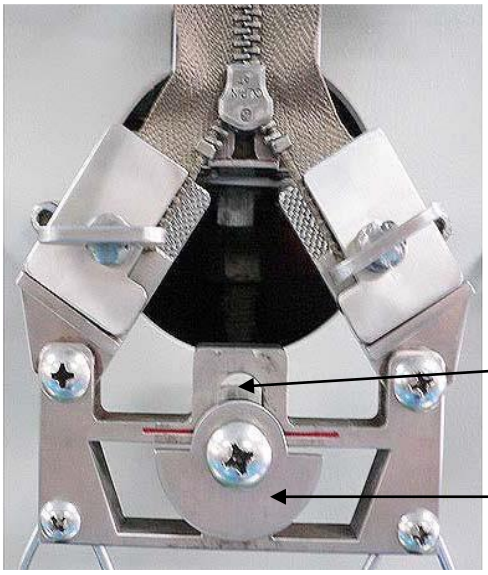


Figure 7

vi). Lock the clamp A. Refer to figure 5.
vii) Adjust the spring balances E (See figure 9) to apply the appropriate longitudinal forces.
Viii) Fix the fastener tapes to the lateral clamps F, leaving gaps of 5mm between the lateral clamps and the chain at the nearest points (the gaps can be gotten by adjusting the screw H.), namely clamp at the reference lines (Figure 1). Ensure that the center lines of the lateral clamps F coincide with the strings that pass over the pulleys. (Please skip this step if a short zip fastener is used)

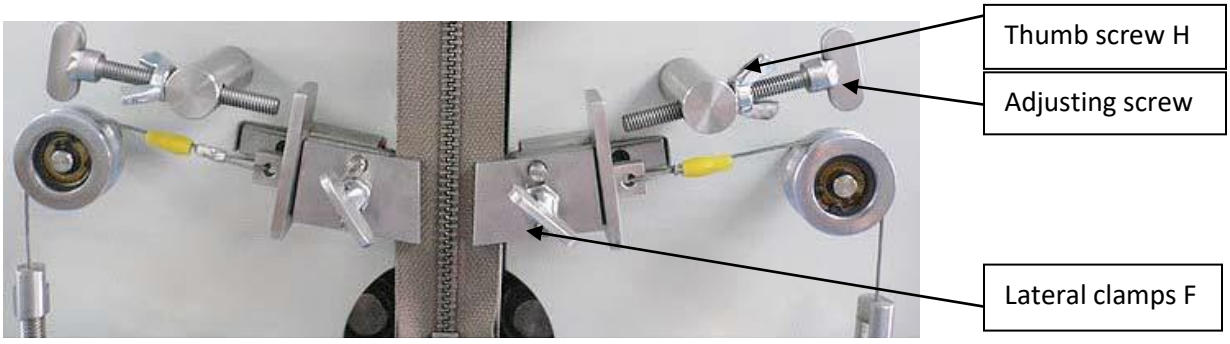


Figure 8

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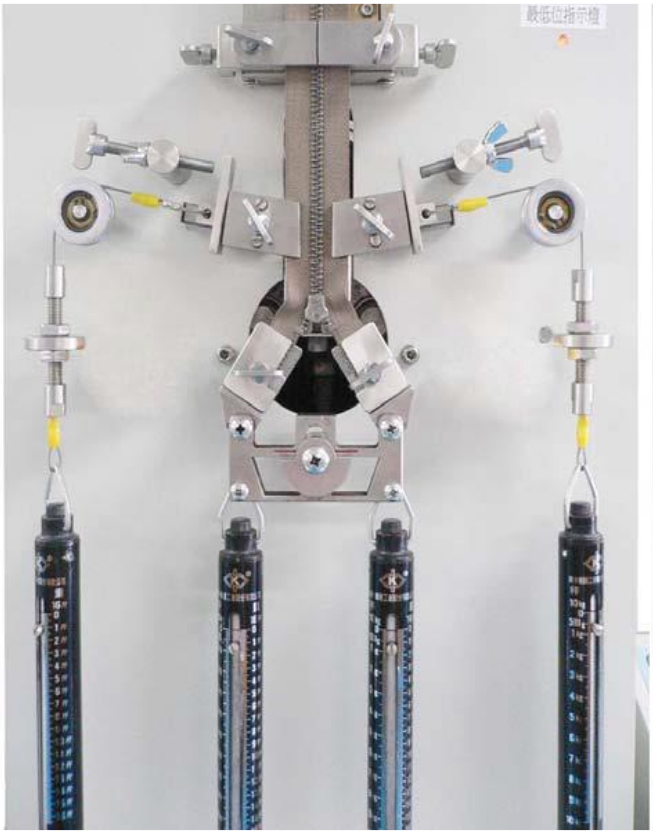
NOTE: Ensure that the lateral clamps F will not clamp elements and will clamp the specimen at the same positions. The vertical lines of the lateral clamps F shall be vertical to the edges of the elements.

ix) Adjust the spring balances G (See figure 9) to apply the appropriate lateral forces. (Please skip this step if a short zip fastener is used)

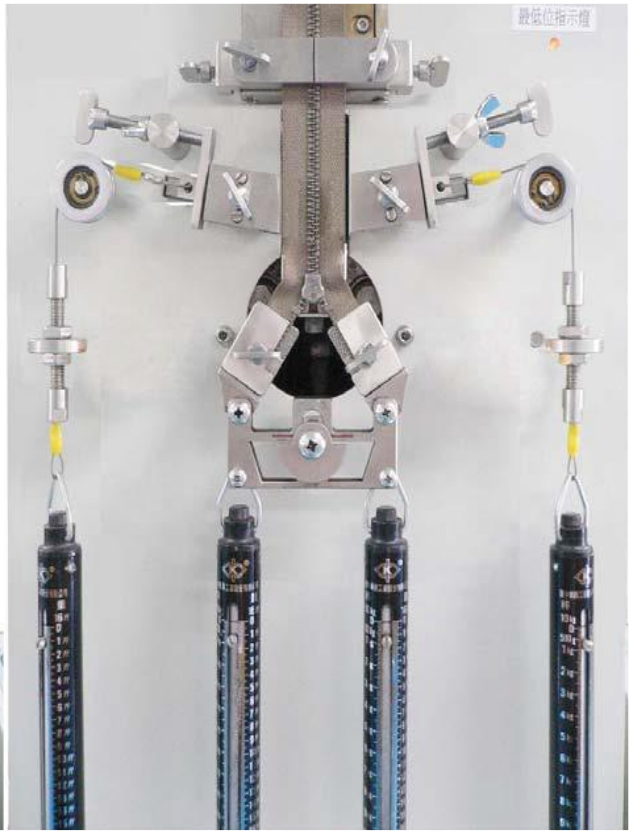
x) Set a required test value on the control panel (refer to the control panel section for more details), and then start the machine to allow its clamping devices to operate for on cycle to equalize the position of specimen.

Caution:
Check the forces of spring balances E and G, and adjust when necessary.When the slider clamping device (See figure 4) is in the highest position, you can use the 30° angle gauge to measure the angle α when the specimen is opened.

Figure 9



Completed installation of long specimen



Completed installation of short specimen

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Note: In accordance with the type of the slide fastener, adjust respectively the top and bottom nuts of the lateral spring balances and the longitudinal spring balances to apply appropriate lateral and longitudinal forces to the stringers as shown in table 1

Table 1

Forces (F1 & F2) vs Element				
Type of Element or Teeth	**Type of Shoes**	**Fastener Performance**	**F1 - Springs E Longitudinal Force (kgf)**	**F2 - Springs G Lateral Force (kgf)**
Plastic	Kids, Children &Casual	Light	0.90	1.00
Metal	Kids, Children &Casual	Medium	1.40	1.60
Metal	Athletic	Medium-Heavy	1.90	2.50
Metal	Performance	Heavy	2.50	3.00

3. Set the speed at 30 ± 3 cycles per minute

4 Set the specified number of cycles on the control panel and clear any figure seen on the actual test value.

5. At the end of the specified number of cycles, remove the specimen and turn off the power switch.

Note: While the test is ongoing don't adjust F1- springs E longitudinal force and F2-springs G lateral force.

6. After completion of the test, remove the specimen and check for any damage.
   - Detachment of slider from one track.
   - Breakage of the puller.
   - Jamming or wear of the element / teeth
   - Detachment or breakage of the teeth
   - Stitch abrasion (sewn -on nylon chain)
   - Tearing of stringer tape.

REPORT: Number of cycles the failure was noticed.